

HOTEL RESERVATION ANALYSIS PROJECT REPORT

TABLE OF CONTENTS

EXECUTIVE SUMMARY	3
INTRODUCTION	4
DATA OVERVIEW	4
METHODOLOGY	5
ANALYSIS AND INSIGHTS	7
RECOMMENDATIONS	15

EXECUTIVE SUMMARY

This report provides a comprehensive analysis of hotel booking data from 2015 to 2017, sourced through the Data Science Apprenticeship Program. The objective was to uncover insights into booking behaviors, cancellation patterns, and customer preferences, which can guide strategic decision-making for hotel management.

Our findings reveal significant trends in cancellation rates, which correlate with lead times and previous cancellations, suggesting areas for policy adjustment to enhance booking retention. We also identified seasonal variations in bookings and cancellation rates, highlighting opportunities for targeted marketing and dynamic pricing strategies to maximize revenue during peak and off-peak seasons.

Through rigorous data cleaning, analysis using SQL, and visualization with Tableau, this report provides actionable recommendations aimed at improving customer engagement, optimizing booking processes, and reducing cancellations. The suggested strategies are geared towards enhancing overall operational efficiency and customer satisfaction in a competitive hospitality environment.

INTRODUCTION

In the competitive landscape of the hospitality industry, understanding customer behavior and booking patterns is crucial for optimizing operational efficiency and enhancing guest satisfaction. This project aims to dissect and analyze a dataset that contain information about reservation from two types of hotels to uncover insights that can drive strategic decisions and improve business outcomes.

The primary goal of this analysis is to identify trends such as; determining factors influencing booking cancellations, understanding the distribution of guest demographics across various hotels, among others. By leveraging data-driven insights, the hotels can tailor its marketing strategies and improve service delivery to better meet the needs of its customers.

This report will detail the methodology used for analyzing the dataset, present the findings from the exploratory data analysis, and offer actionable insights and recommendations based on the results. Through this thorough analysis, (the project seeks to empower the hotels with the knowledge to enhance its booking strategies and ultimately boost its revenue and customer satisfaction metrics).

DATA OVERVIEW

The dataset for this analysis was procured through the FLIT Data Science Apprenticeship Program. It comprises two primary components: the first sheet outlines the column descriptions and provides a metadata framework, while the second sheet contains the actual data, rich with detailed information on reservations, bookings, guests, and other related metrics.

Initially, the dataset encompassed 119,390 records and 32 distinct columns, spanning three years from 2015 to 2017. This extensive dataset covers a global perspective, offering insights into hotel booking dynamics worldwide. The comprehensive nature of the data allows for a multifaceted analysis of trends and patterns in hotel booking behaviors, providing a robust platform for developing actionable insights in the hospitality domain.

This overview sets the stage for a deeper exploration into the specific methodologies employed to prepare and analyze the data, ensuring that subsequent findings are built on a well-understood and rigorously prepared dataset.

METHODOLOGY

Data Preparation and Cleaning: The primary dataset used in this analysis was obtained from the Data Science Apprenticeship Program. Initially, the dataset comprised 119,390 rows and 32 columns, representing reservation details from 2015 to 2017. Data cleaning was a crucial initial step in the analysis process to ensure the accuracy and relevance of the findings. This step involved:

- **Handling Missing Values:** Rows with missing data (approximately 400 entries) were evaluated for significance and potential bias before deciding on imputation or removal. columns 'agent' an 'company' was dropped for the high volumn of missing data. company column has over 94% of the data missing while agent columns had over 13% of data missing. column 'children' had 4 rows with missing data. This was filled with the mode: 0. Column 'country' had 488 rows with missing values, which were all dropped. The high volume of data allowed for the removal of these rows without significant loss of information.
- **Outlier Detection and Treatment:** Outliers, such as an 'ADR' (Average Daily Rate) of 5400 and rows with 10 babies, which is significantly higher than the norm, were identified and investigated. Outliers were treated based on their deviation from the mean and their potential impact on overall analysis, with these values being removed to prevent distortion in the analysis.
- **Duplicate Value Treatment:** During the initial data preparation phase, we identified 31,994 duplicate rows within the dataset. These duplicates were exact replicas in terms of booking details, guest information, and reservation timelines, suggesting errors in data entry or integration. To maintain the integrity of our analysis and ensure accuracy in our findings, all identified duplicate entries were removed from the dataset. This step was crucial for preventing any bias that could arise from overrepresented data and for ensuring the reliability of subsequent statistical analysis.
- **Data consistency:** Consistency: Five distinct meal type were found, which was inconsistent with the original column descriptions of four types. The values were standardized to ensure consistency across the dataset.

Post-Cleaning Data Overview: After the validation and cleaning process, the dataset remained with 84926 rows and 30 columns.

Data Transformation: Critical data transformations were implemented to enhance the analysis framework and data usability. Noteworthy transformations included:

- **Conversion of Data Type:** Certain columns required data type changes to facilitate analysis. For example, date fields initially stored as strings were converted to datetime objects to enable time series analysis.

- **Creation of New Variables:** Derived variables were introduced to better capture insights. Notable examples include calculating the 'Total Stay Duration' from summing total weekend and week nights spent and 'Booking date' by subtracting the lead time from the arrival date. These transformations allowed for more nuanced segmentation and trend analysis.

These transformations ensured that the dataset was optimized for analytical tasks, enhancing both the performance of analytical models and the clarity of the insights derived.

Validation Procedures: Throughout the data cleaning and transformation phases, validation checks were performed to ensure that data manipulation retained the integrity of the dataset. This included spot checks, statistical summaries, and visualization of key metrics to confirm consistency and reasonableness.

Tools and Software Used: The data was processed using SQL for robust data querying and manipulation, which was essential for handling the dataset's complexity and size. For visualizations, Tableau was employed to create interactive dashboards and charts, providing clear and effective communication of findings. These tools were chosen for their powerful capabilities in data analysis and user-friendly visualization features, ensuring an efficient and comprehensive exploration of the data.

ANALYSIS AND INSIGHTS

This section explores the dynamic patterns of hotel bookings, cancellation rates, and guest preferences over various periods and demographics. The insights derived from a comprehensive suite of visualizations help to understand factors influencing customer behavior and hotel performance, aiming to optimize operational and marketing strategies.

Booking Analysis

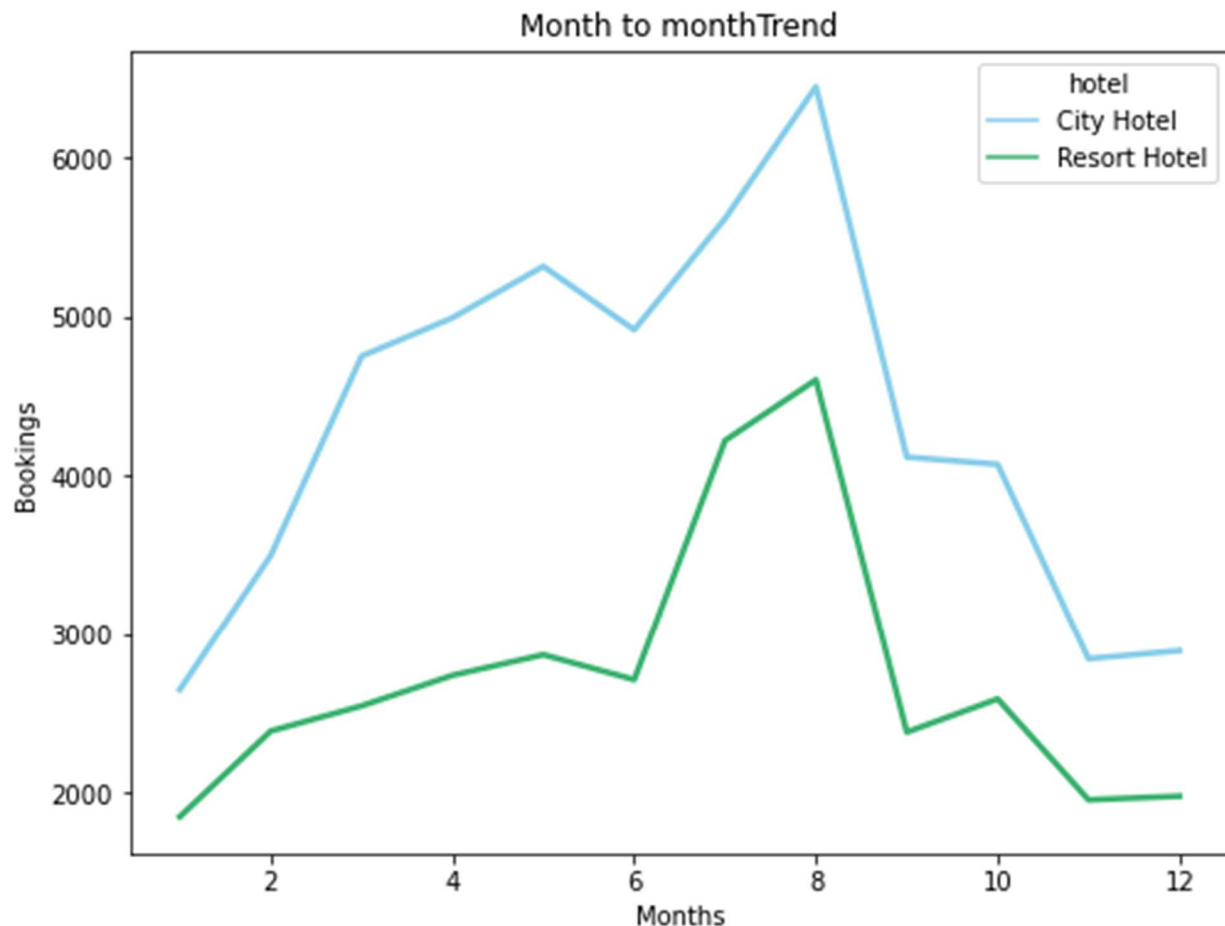
Figure 1: Total Bookings by Hotel

The bar chart comparing total bookings between City Hotel and Resort Hotel clearly illustrates that City Hotel has secured more bookings than Resort Hotel. This visualization effectively highlights the disparity in guest preference or demand between the two establishments.



Figure 2: Monthly booking trends

The line chart (Figure 2) below reveals the fluctuation in the number of bookings per month throughout the years. A significant peak is observed in August, with over 6000 bookings for city hotel and 4601 bookings for Resort Hotel, corresponding with the summer vacation season. Conversely, the lowest booking month is November, with under 100 bookings, indicating a possible off-peak season for the hotels. This pattern suggests a strong seasonal influence on booking behavior, crucial for planning promotions and staffing.



Cancellation Analysis

Figure 3: Cancellation rate by hotel

The pie chart illustrates the cancellation rates between the two hotels, with the City Hotel cancellation rate being nearly double that of the Resort Hotel. This stark contrast highlights significant differences in cancellation behaviors at each location.

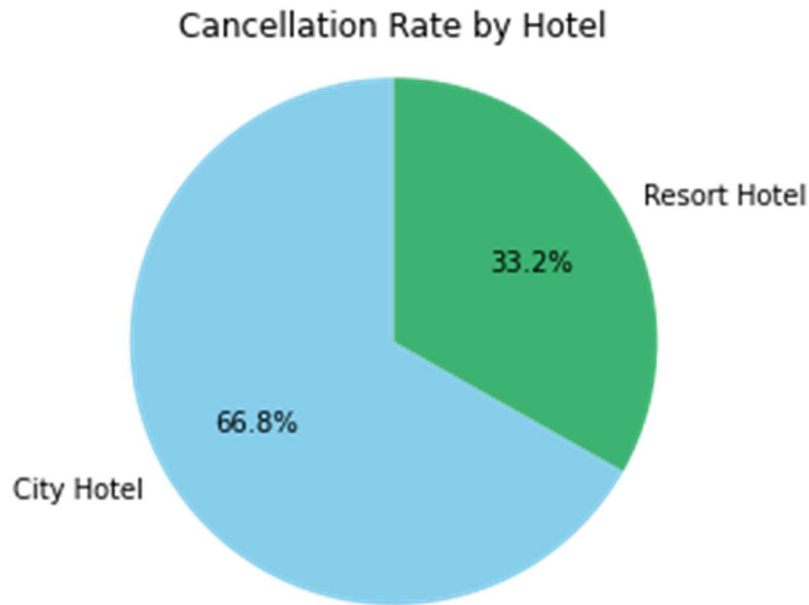


Figure 4: Cancellation Rates as a Function of Lead Time

Figure 4 presents a scatter plot that maps cancellation rates against lead times. There is a discernible trend where longer lead times are associated with higher cancellation rates. Specifically, bookings made more than 100 days in advance tends to result in cancellation. This finding could inform cancellation policy adjustments to mitigate loss from early bookings

A direct relationship between cancellation rates and lead times has been established, indicating potential revenue losses with longer lead times.

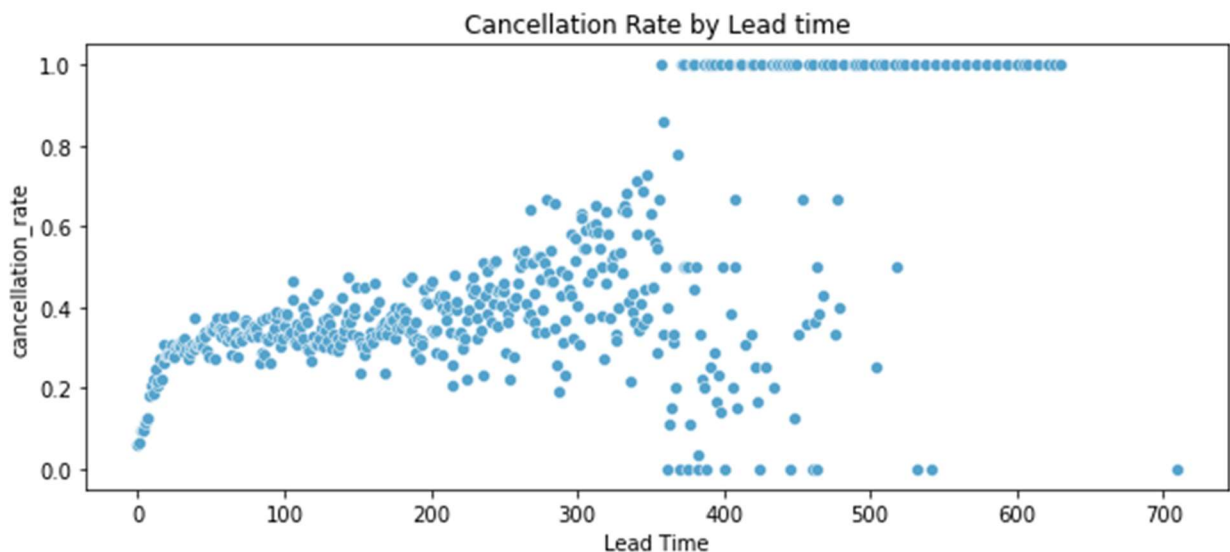
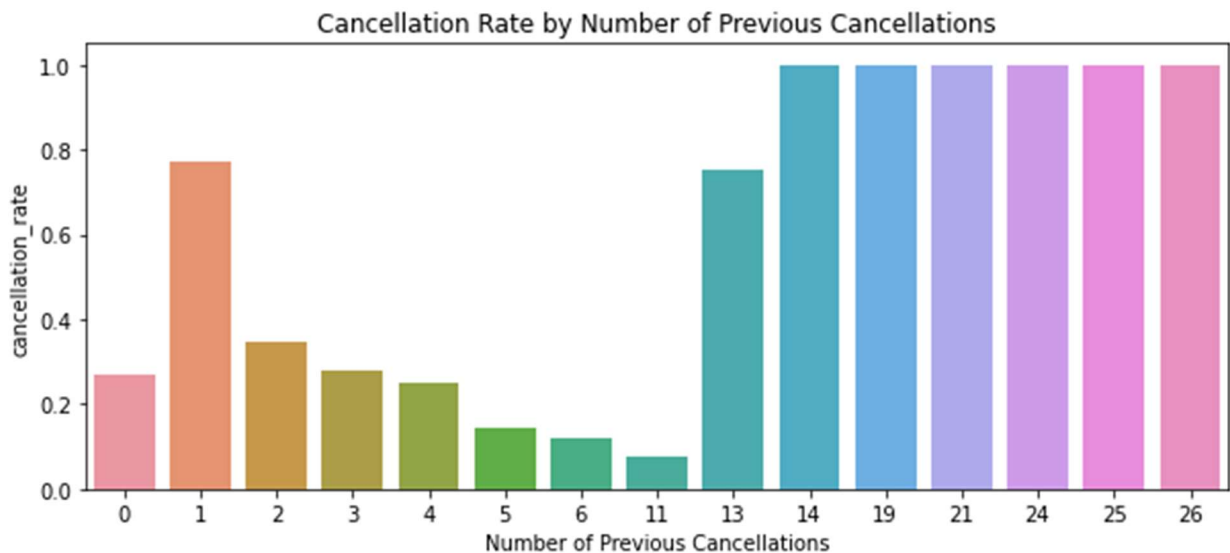


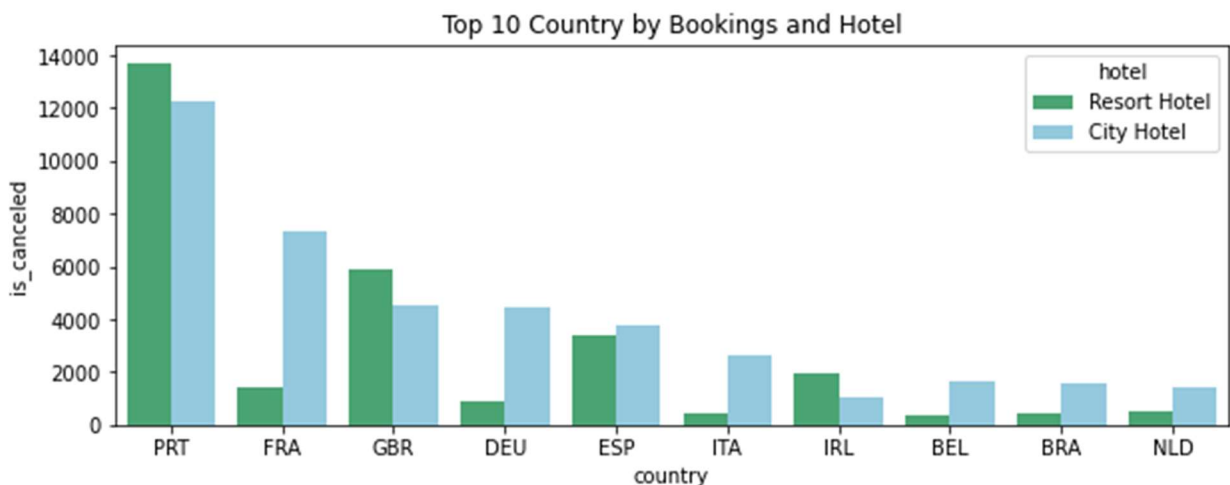
Figure 5: Cancellation Rate by Number of previous cancellations

The chart shows a bar chart that highlights the relationship between previous cancellations and cancellation rates. The findings indicate a clear trend: a higher number of previous cancellations correlates with increased future cancellations. This pattern indicates a high risk of repeat cancellations among guests who have canceled before. Interestingly, even guests who had only one previous cancellation exhibited a significantly elevated cancellation rate. This suggests that identifying guests with any cancellation history could be key to reducing future cancellations.



Guest geographical distribution

Figure 6: Geographic Analysis of Guest Origins This bar chart illustrates the top 10 countries from which the majority of guests originate. The Portugal and France lead, highlighting the local and regional market's importance. Understanding these geographical patterns helps tailor marketing campaigns to specific demographic segments.



Customer Type Analysis

Figure 7: Customer Types by Country

A side bar chart (Figure 7) displays the proportions of different customer types within the top ten guest-origin countries. This visualization underscores the diversity in guest types in the different countries. Such insights are valuable for customizing service offerings to cater to varying customer needs.

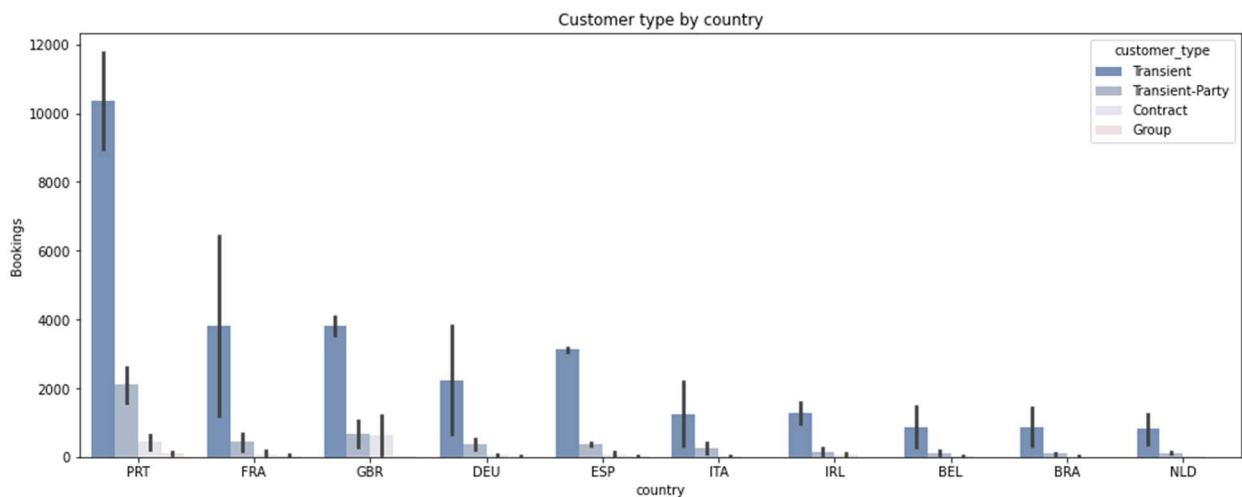


Figure 8: Most popular rooms

The bar chart displays the popularity of reserved room types across the hotels. It clearly shows which room types are most frequently booked, providing insights into guest preferences and potential areas for room upgrades or promotions.

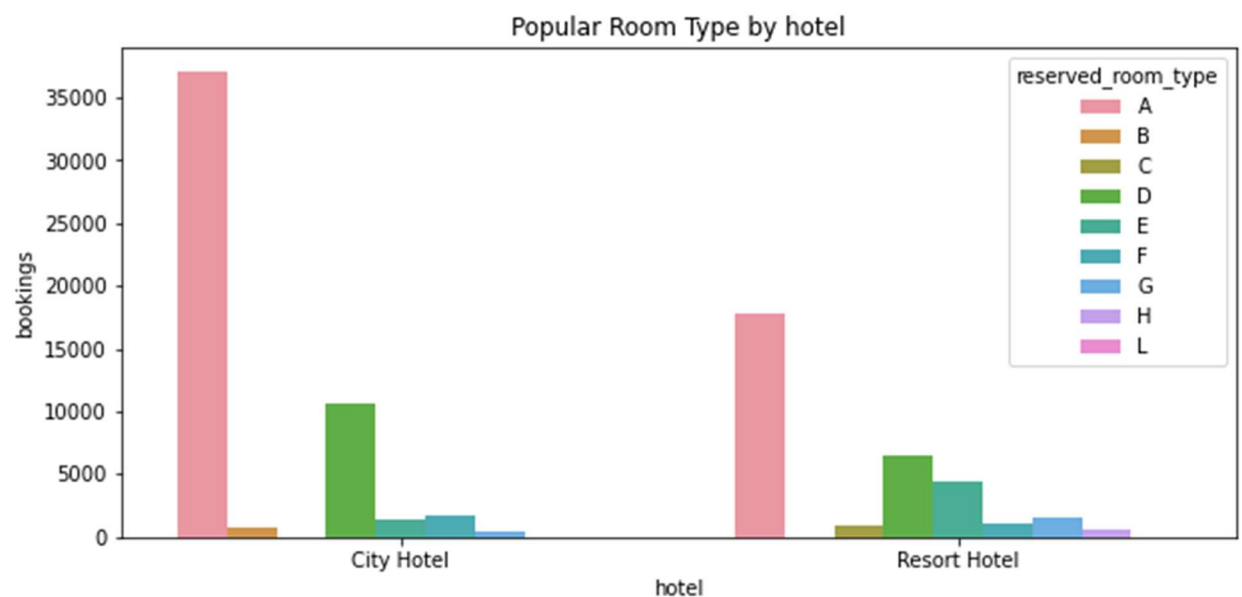
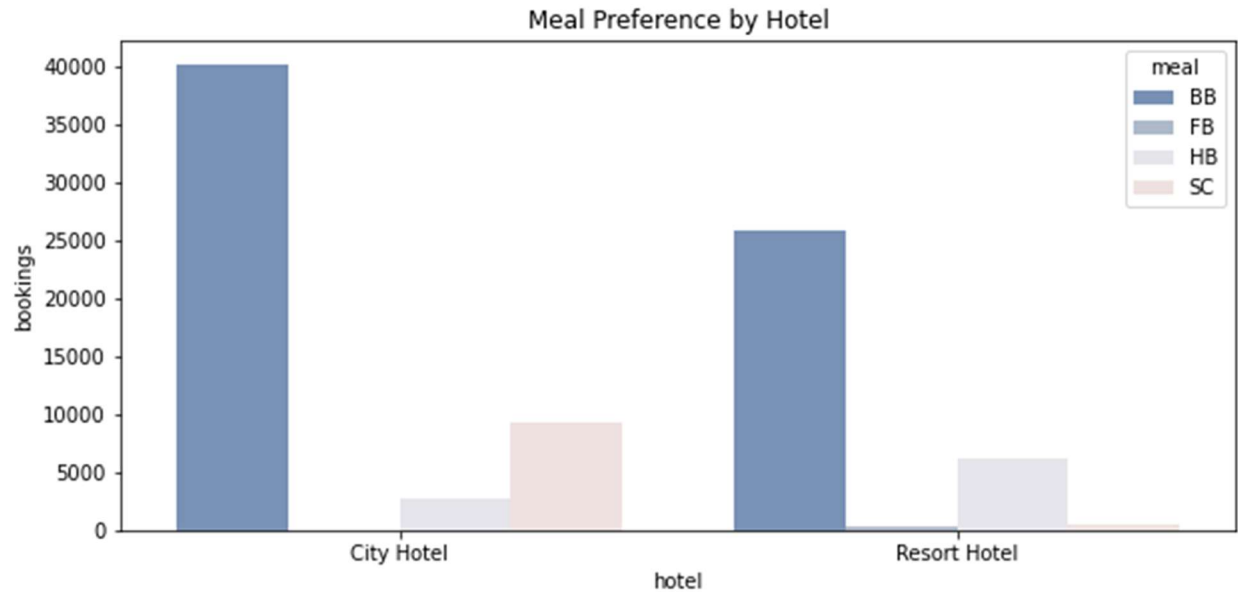


Figure 9: Meal preferences of the guests

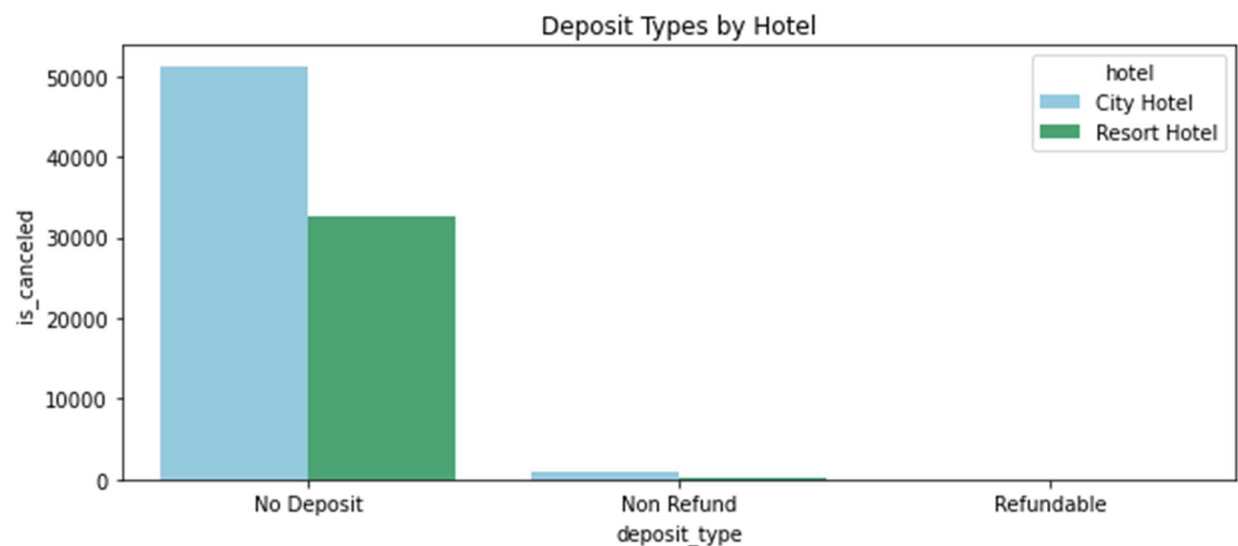
The bar chart shows the popularity of meal types across the hotels. With meal type Bed and Breakfast (BB) being the meal guests in both hotels prefer.



Deposit Analysis

Figure 10: Deposit Analysis by hotel

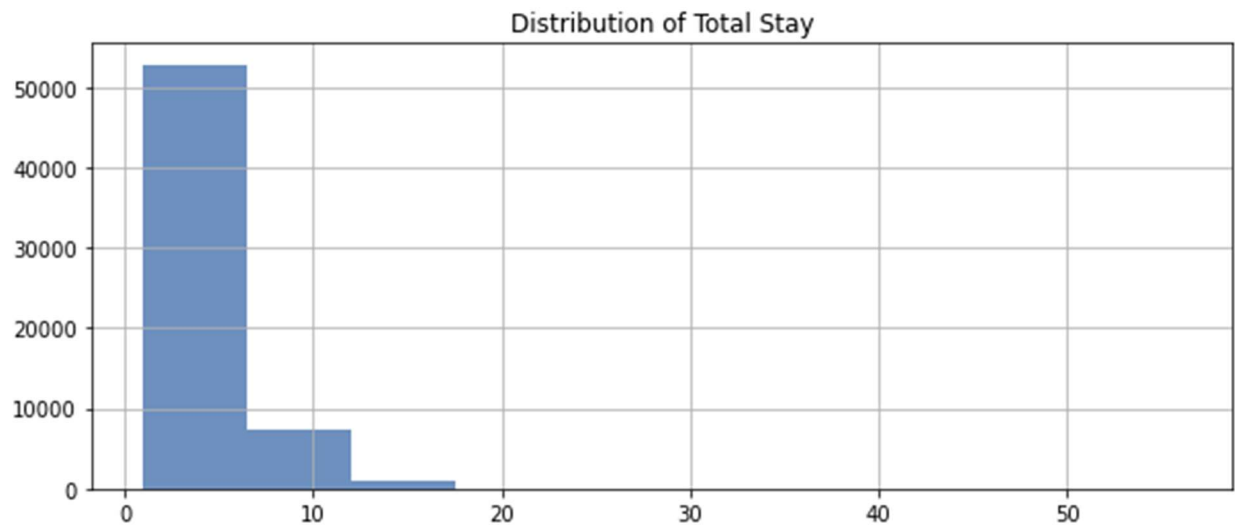
The bar chart illustrates the proportion of bookings involving a deposit for different types of hotels. It highlights variations in deposit policies between hotel types, suggesting potential strategies for managing financial risks and ensuring revenue stability.



Length of stay distribution

Figure 11: Distribution of Stay Lengths

A histogram of stay lengths (Figure 11) shows that the majority of stays are between one and four nights, with a mode at two nights. This distribution is indicative of a trend towards short stays, possibly driven by weekend getaways or short business trips.



Adr Analysis

Figure 12: Adr by customer type

The bar chart compares the average daily rate (ADR) by customer type across different hotels. It clearly demonstrates how ADR varies among customer segments, showing the types of customers that contribute most to revenue more, potentially guiding targeted marketing strategies.

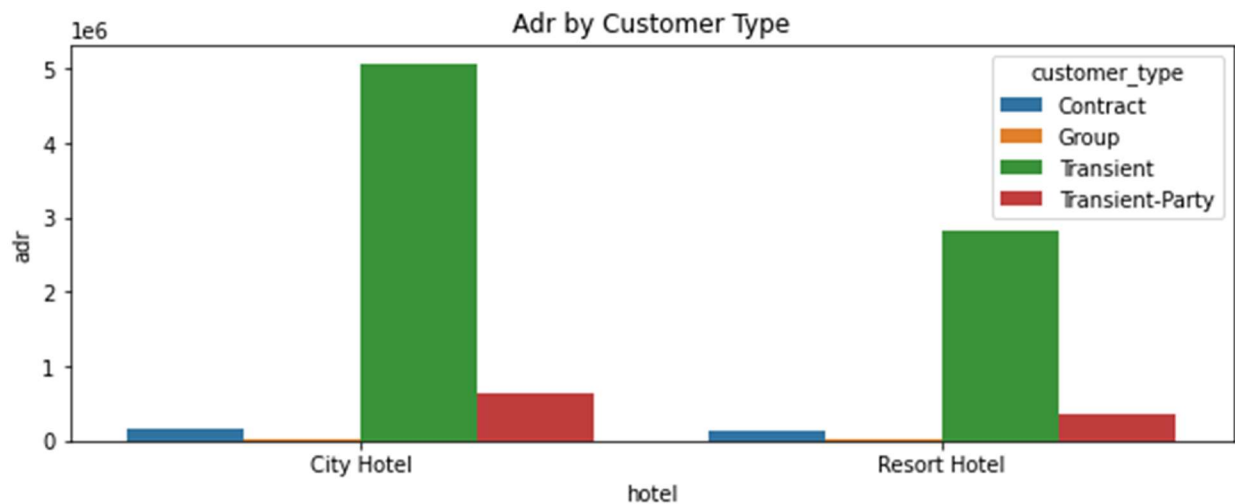
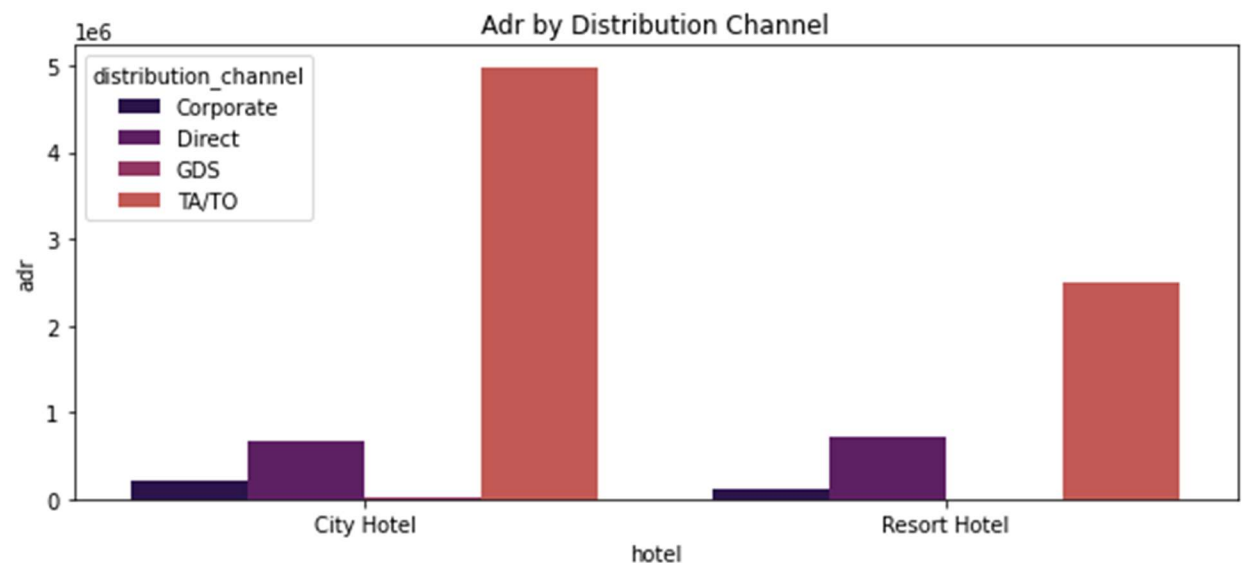


Figure 13: ADR by distribution channel

The bar chart illustrates the average daily rate (ADR) distribution by channel for each hotel, highlighting significant differences in pricing strategies and guest acquisition costs across various booking platforms. This visualization can help hotel managers identify the most lucrative channels and optimize their distribution strategies accordingly.



Conclusion of Findings

The analyses conducted reveal clear patterns in booking behavior, cancellation rates, guest demographics, and stay preferences. Seasonal trends indicate the need for targeted operational planning, while the varied preferences across different guest origins and types suggest opportunities for personalized marketing and service offerings. The insights from these findings will guide strategic decisions to enhance guest satisfaction and optimize revenue management.

RECOMMENDATIONS

Based on our detailed analysis of the hotel booking data, several opportunities have been identified to enhance revenue management and improve customer satisfaction. The following recommendations are designed to address the key issues highlighted in the analysis.

- 1. Enhance Customer Loyalty Programs:** Develop personalized loyalty programs to increase retention rates, particularly since over 90% of bookings are from first-time guests. Analyze guest booking behaviors and spending patterns to offer customized rewards, promoting these programs through targeted emails and social media to enhance guest retention.
- 2. Optimize Distribution Channels:** Focus on channels with the highest average daily rates (ADR). Assess the cost-effectiveness of each channel and reallocate marketing budgets to maximize revenue while minimizing acquisition costs.
- 3. Implement Strategic Pricing During Peak Cancellation Periods:** Introduce dynamic pricing strategies to offer discounts during months with historically high cancellation rates. This approach aims to reduce cancellations and stabilize occupancy rates by incentivizing early bookings and managing risks associated with longer lead times.

CONCLUSION

This analysis of booking data has illuminated critical operational dynamics and customer behaviors impacting profitability and satisfaction. Implementing strategic recommendations such as personalized loyalty programs, optimized distribution channel management, and dynamic pricing during peak cancellation periods will empower the hotel to capitalize on these insights. Such strategic initiatives are expected to enhance revenue, improve customer retention, and ensure competitive advantage in the hospitality industry.